PollisDatasets.jl: the datasets of the examples

67 small datasets, from `mtcars` and `iris` to `mroz` and `challenger`, each with the source to cite, its licence and every column described, loaded as Tables.jl column tables.

In Pollis	the data of the Code Examples, installed with Pollis; the database *PollisDatasets* in the Databases view, the `duckdb>` terminal and the `sql>` mode, one table per dataset, with its descriptions.
Missing in Julia	a set of datasets whose licences were checked one by one, and recorded, before they came in. RDatasets.jl loads the datasets of R's packages, as DataFrames, without the terms of each.
Package	`PollisDatasets.jl` 0.1.0, from GitHub: `Pkg.add(url = "https://github.com/Trumpingtons/PollisDatasets.jl")`. It depends on CSV, TOML and Tables only.
Tests	68 of 68: every dataset loads with the rows and the columns its entry gives, and an unknown name is an error. Six datasets also checked against R's datasets package, to the last digit printed.

Why a package

An example is only worth running if its data comes with it. Pollis's Code Examples need datasets that are small, well known and free to redistribute with an AGPL-3.0 program; a teacher or a student needs to know where each one comes from and how to cite it. PollisDatasets.jl puts each dataset in `data/<id>.csv`, with its entry in `datasets.toml`: a title, a domain, a description, the number of rows, the source to cite, where the file was taken from, the licence the provider states, the outcome of the licence check, and a description of every column, with its units or codes. Missing values are empty fields in the file and `missing` in Julia.

What it holds

Domain	Datasets
Regression	mtcars, anscombe, datasaurus, galton, evals
Econometrics	mroz, card
Panel econometrics	grunfeld, produc, gasoline, petersen
Causal inference	lalonde, cps1, cps2, cps3, mpdta, divorce
Count data	horsekicks, insectsprays, biochemists, ships, challenger, assortive_mating
Classification	penguins, iris, titanic, wine, breastwisconsin, glass, sonar, vowel, ionosphere, germancredit, image, ecoli, votes, letters, satimage, vehicle, liver, soybean, pima
Time series	airpassengers, nile, co2, bg96, dgs10, lynxmackenzie, airquality, sp500, gamestop
Time series classification	ecg200, earthquakes, twoleadecg
Survival	lung, veteran, pbc
Bayesian / multilevel	eightschools, radon, sleepstudy
Dynamical systems	lynxhare
Epidemiology	italycovid
Density / mixtures	faithful
Market baskets	groceries
Text	pollis_guide, the sections of The Pollis Guide
Simulated	sim_glmm, sim_clf

The licences

Every dataset passed a licence check before it came in, and its `status` records the outcome:

Status	Meaning	Datasets
`open`	an explicit open licence (CC0, CC BY), or permission given	23
`public-domain`	a work of the US government, or published before 1930	5
`redistributed`	no explicit licence; a small factual table redistributed for decades in CRAN and textbook packages	37
`simulated`	made by the Julia code in the entry's `code`, which states the true model; MIT, like the package	2

The datasets that failed are listed at the top of `datasets.toml` with the reason: the Fama-French factors (no grant to

redistribute: fetch them live, from **Explore › Access Data Libraries**), the SILSO sunspots (CC BY-NC, whose non-commercial clause AGPL-3.0 cannot carry) and Boston Housing (an ethically problematic variable). The S&P 500 series came in only as a copy dedicated to the public domain (CC0), with the source named.

In Julia

Function	Returns
`datasets()`	the catalogue, one row per dataset: `id`, `title`, `domain`, `rows`
`dataset("mroz")`	the dataset as a Tables.jl column table, a `NamedTuple` of vectors: `d.wage` is a column
`datasetinfo("mroz")`	the description, the source to cite, the licence and the columns

A column table goes straight into GLM.jl, MLJ.jl, Makie or any Tables.jl reader; Pollis's examples never turn it into a DataFrame.

In Pollis

- **Installed with Pollis.** When the Julia REPL starts, Pollis adds PollisDatasets.jl to an environment of its own, `~/.pollis/julia/environments`, on the load path, so `using PollisDatasets` works in any project without touching its `Project.toml`, and updates it when GitHub has a newer commit.
- **The database.** `PollisDatasets.duckdb`, in `~/.pollis/databases`, has one table per dataset, named by its id, with the title, description, source and licence as the table's comment and each column's description as its own. It is attached read-only to every connection Pollis makes, so `SELECT * FROM PollisDatasets.mtcars` works in the `sql>` mode and the `duckdb>` terminal, and the descriptions show in the Databases view. It is rebuilt when the installed package or its files change.
- **New datasets announced.** When a version brings new datasets, the Welcome page names them, with **Open Databases View**.

A worked example

Anscombe's quartet

```
using PollisDatasets, Statistics
datasetinfo("anscombe")                          # 11 rows; Anscombe (1973), The American Statistician
a = dataset("anscombe")
x, y = a.x1, a.y1                                                          # and x2, y2 to x4, y4
b = cov(x, y) / var(x); (mean(y) - b * mean(x), b, cor(x, y))
```

Four sets of eleven points, built by Anscombe (1973) to show why a regression should be drawn before it is believed: the same means, the same variances, the same correlation and the same fitted line, from a straight line with noise, a curve, a line with one outlier, and a vertical cluster with one far point.

Set	mean x	var x	mean y	var y	cor	intercept	slope
1	9	11	7.500909	4.127269	0.816421	3.000091	0.500091
2	9	11	7.500909	4.127629	0.816237	3.000909	0.500000
3	9	11	7.500000	4.122620	0.816287	3.002455	0.499727
4	9	11	7.500909	4.123249	0.816521	3.001727	0.499909

The values are Julia's on `dataset("anscombe")`; R's `lm` on its own `anscombe` gives the same to six decimals.

What the tests show

The catalogue and the files agree

`test/runtests.jl` loads every dataset in the catalogue and checks that it has the number of rows its entry gives and the columns its entry describes, in the same order; and that `dataset("nope")` is an error that lists the names there are. 68 of 68 pass, in 7 seconds.

Against R

Datasets that come from R's datasets package were compared with R 4.5's own copies:

Dataset	Column	Rows	Values	Sum	Mean
mtcars	mpg	32	32	642.9	20.09062
iris	sepal_length	150	150	876.5	5.843333
faithful	eruptions	272	272	948.677	3.487783
airquality	ozone	153	116	4887	42.12931
nile	flow	100	100	91935	919.35
airpassengers	passengers	144	144	40363	280.2986

Both give these values, R to the digits it prints; the 37 ozone readings missing in New York's summer of 1973 are `missing` in Julia as they are `NA` in R.

Adding a dataset

In the GitHub clone of PollisDatasets.jl, never in the installed copy: check the licence and choose the `status`; add `data/<id>.csv` and its entry in `datasets.toml`, with every column described; run the tests; commit and push. At the next start of the Julia REPL, every Pollis installs the new version, rebuilds the database and announces the dataset on the Welcome page. **Help** › **Documentation** › **Drill-down: Data Packages** follows the whole road.

References

Anscombe, F. J. (1973). Graphs in statistical analysis. The American Statistician 27(1), 17-21. doi:10.1080/00031305.1973.10478966

R Core Team. The R datasets package. `https://stat.ethz.ch/R-manual/R-devel/library/datasets/html/00Index.html`

Software: CSV.jl, Tables.jl and TOML (MIT). The source of each dataset is in its entry of `datasets.toml`.

Package written and tested with Claude Code (Anthropic) in Pollis.

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